IGCSE ECONOMICS · SECTIONS 1.1 TO 1.4

# Study Companion: The Basic Economic Problem

Scarcity, factors of production, opportunity cost and production possibility curves. Learn the key words, avoid the common traps, practise multiple choice, then build full Paper 2 answers step by step. Model answers and student mark schemes are at the back, so you can mark your own work.


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Name: _____ Class: _____

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# The big idea

All four sections in this book are one story. Learn the story first, then the details.

**1.1 Scarcity**
Wants are unlimited, but resources are finite.

→

**1.2 Resources**
The resources are the four factors of production: land, labour, capital, enterprise.

→

**1.3 Choice**
We must choose. Every choice has an opportunity cost.

→

**1.4 The PPC**
A diagram that shows scarcity, choice and opportunity cost.

Khan hiếm → nguồn lực (yếu tố sản xuất) → lựa chọn và chi phí cơ hội → đường giới hạn khả năng sản xuất (PPC).

# Paper 2 toolkit

Every Paper 2 question in this guide has three parts, like the questions in your exam: **(a) 2 marks, (b) 4 marks, (c) 6 marks = 12 marks**. The command word tells you what kind of answer to write.

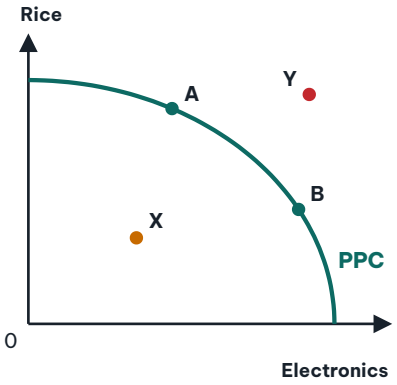
| Command word | What it means | How to build your answer | Time |
|---|---|---|---|
| **Define** / **State** [2] | Give the exact meaning, or give a short fact. | One sentence with the **two key ideas** of the definition. No example needed. | 3 min |
| **Explain** [4] | Say **what** something is and **why** or **how**. | Two points. For each: **make the point** (1 mark), then **develop it or give an example from the context** (1 mark). | 6 min |
| **Analyse** [6] | Show the steps from cause to effect. | A **chain of reasoning**: each link is one step. Use "so", "this means", "as a result". Add a labelled diagram when it helps. | 9 min |

## How to write a chain of reasoning

Examiners give marks for each **link** in the chain. Do not jump from the start to the end. A Unit 1 chain often looks like this:

**Change** → **Factors of production (quantity or quality)** → **Productive capacity** →
**Output** → **PPC shifts or point moves** → **Opportunity cost / effect on people**

Chuỗi lập luận: thay đổi → yếu tố sản xuất → năng lực sản xuất → sản lượng → PPC dịch chuyển hoặc điểm di chuyển → chi phí cơ hội / tác động.




A complete PPC diagram.

### PPC diagram checklist

☐ Both axes labelled with **two goods** (for example Rice and Electronics). **Never** Price and Quantity.

☐ The curve is **bowed out** from the origin, and it touches both axes

☐ The curve is labelled PPC (or PPC1 and PPC2)

☐ Points are labelled with letters

☐ An arrow shows a movement along or a shift

☐ Dotted lines to the axes when you show opportunity cost

## How to mark your own work

1. Finish the whole answer first, then use a different colour pen.
2. Use the mark scheme table at the back. Tick each row only if the idea is **clearly** there. Different words are fine if the meaning is the same.
3. Write the mark next to each tick, then the total. Do not go over the maximum.
4. Compare with the model answer. Write one thing you did well and one step that is missing, then rewrite that step.

# 1.1 SECTION 1.1
# The basic economic problem

**In this section:** 1.1.1 finite resources and unlimited wants · 1.1.2 resource allocation: what, how and for whom · 1.1.3 economic goods and free goods

## Key words

### The basic economic problem
How to use finite (limited) resources to satisfy unlimited wants.

**Simply:** we want more than we can make, so we must choose.

**vấn đề kinh tế cơ bản**

### Scarcity
When there are not enough resources to satisfy everyone's wants, because wants are unlimited and resources are finite.

**Simply:** there is not enough for everything we want.

**sự khan hiếm**

### Finite resources
Resources that are limited in amount at any one time: land, labour, capital and enterprise.

**Simply:** there is only so much land, so many workers and so many machines.

**nguồn lực hữu hạn**

### Unlimited wants
People's wants never end. When one want is satisfied, new wants appear.

**Simply:** after a new phone, you want a new laptop.

**nhu cầu (mong muốn) vô hạn**

### Economic agents
The decision-makers in an economy: households (consumers and workers), firms (producers) and the government.

**Simply:** everyone who makes economic choices faces scarcity.

**các chủ thể kinh tế (hộ gia đình, doanh nghiệp, chính phủ)**

### Resource allocation
Deciding how scarce resources are used, by answering three questions: **what** to produce, **how** to produce it and **for whom**.

**Simply:** every economy must make these three choices.

**phân bổ nguồn lực: sản xuất cái gì, như thế nào, cho ai**

### Economic good

A good that is scarce. Resources are used to make it, so it has an opportunity cost.

**Simply:** making it uses up resources that could make something else.

**hàng hóa kinh tế**

### Free good

A good that is not scarce. Supply is greater than demand at a zero price, and it has no opportunity cost.

**Simply:** there is enough for everyone, and nobody has to give anything up. Examples: air, sunlight, sea water.

**hàng hóa tự do**

### Memory hook: 3 questions every economy must answer

1. **What** to produce? Which goods and services, and how many? *More rice or more phones? A new metro line or a new hospital?*
2. **How** to produce? Which resources and methods? *Robots or workers? Big farms or small farms?*
3. **For whom** to produce? Who gets the output? *Who gets the new flats? Who gets water first in a drought?*

Sản xuất cái gì? Sản xuất như thế nào? Sản xuất cho ai?

## Scarcity affects every economic agent

| Who | What is scarce | Example |
|---|---|---|
| **Consumers** | income and time | Linh has $20. She cannot buy a new game, a T-shirt and a cinema ticket. |
| **Workers** | time and skills | A worker in Bình Dương cannot work overtime and also take an English course in the same evening. |
| **Firms** | land, workers, machines, money | A factory has one production line. It can make phones or TVs on it, not both. |
| **Government** | tax revenue and land | The government cannot build every road, school and hospital that people want. |

**Watch out:** rich people and rich countries face scarcity too. Their wants are still bigger than their resources, and time is always scarce.

**TRAP**

### Free of charge is not the same as a free good

**Wrong:** "A lesson at a state school is a free good, because students do not pay."

**Right:** it is an **economic good**. Teachers, buildings and books are scarce resources, paid for by taxes. They could be used for something else, so there is an opportunity cost. The same is true of free vaccinations and free samples in a shop.

Miễn phí (không phải trả tiền) ≠ hàng hóa tự do.

**TRAP**

### A free good can become an economic good

A good is free only while there is enough for everyone. In the Mekong Delta, river water used to be a free good. Now salt water from the sea makes fresh water **scarce** in the dry season. Families must buy it, and the government must use resources to clean it and pipe it. Fresh water has become an **economic good**.

Other examples: clean air in a polluted city (people buy air purifiers), or sand on a beach that is dug up and sold for building.

## Paper 1 practice: multiple choice

**1. What is the basic economic problem?**

**A** There is not enough money in the economy.

**B** Wants are unlimited but resources are finite.

**C** Prices are too high for many consumers.

**D** Firms produce more than people want to buy.

**2. Which of these is a free good?**

**A** tap water in a Hà Nội flat

**B** a bus ticket in Đà Nẵng

**C** sunlight falling on a rice field

**D** a lesson at a state school

**3. Which question is about *how* to produce?**

**A** Should the factory use robots or workers?

**B** Should the country make more rice or phones?

**C** Who should get the new flats in the city?

**D** How much should a new phone cost?

**4. Why do rich countries also face scarcity?**

**A** They have no free goods at all.

**B** Their governments spend too little.

**C** Their firms cannot make enough profit.

**D** Their wants are still greater than their resources.

## Paper 2 practice: structured question

CONTEXT

### Fresh water in the Mekong Delta

For many years, families in the Mekong Delta took fresh water from rivers and canals for free. Now, in the dry season, salt water from the sea comes far up the rivers. This is called **salt water intrusion**. The river water is too salty to drink or to use on rice fields.

Many families now buy fresh water from trucks. They pay about **$5 for 1,000 litres** (practice figure). Some families collect rain water in the wet season, but it runs out.

The government has a limited budget. It must decide whether to build water reservoirs, pipes or treatment plants, and who should get the water first: homes, hospitals or farms.

**Define** **(a) Define the term *scarcity*.** **[2]**

**To get 2 marks** you need two ideas: what people **want**, and what **resources** are like.

1. **Wants:** *Scarcity is when people's wants are* _____...
2. **Resources:** *...but the resources to satisfy them are* _____

> **Watch out:** "when there is not enough of something" is only half an answer. Say **why**: unlimited wants and finite resources.

Explain

**(b) Explain the difference between an economic good and a free good. Use an example of each from the context.** **[4]**

**To get 4 marks:** what an economic good is (1) + example from the context (1); what a free good is (1) + example from the context (1).

1. **Economic good:** *An economic good is _____, so making it has an _____.*
2. **Example:** *For example, _____.*
3. **Free good:** *In contrast, a free good is not scarce: _____, so it has no opportunity cost.*
4. **Example:** *For example, _____.*

**Word bank:** scarce | opportunity cost | uses resources | supply is greater than demand at a zero price | water from trucks | river water in the past

**Watch out:** the context shows that river water **was** a free good and fresh water **is now** an economic good. Saying this clearly shows the examiner you understand the idea.

**Analyse** **(c) Analyse why scarcity means the government must decide what to produce, how to produce and for whom to produce.** **[6]**

**To get 6 marks:** six links in a chain. Start with scarcity, then use **all three** questions, then finish with opportunity cost.

1. **Scarcity:** *Fresh water and the government's budget are _____, but people's wants are unlimited.*
2. **Choice:** *So the government cannot satisfy all wants. It must choose how to _____ its scarce resources.*
3. **What:** *It must decide what to produce, for example _____.*
4. **How:** *It must decide how to produce it, for example _____.*
5. **For whom:** *It must decide for whom, for example _____.*
6. **Opportunity cost:** *Every choice has an opportunity cost: money spent on _____ cannot be spent on _____.*

> **Watch out:** use the context in every step. "The government must decide what to produce" with no example from the water story is a weak link.

# 1.2 SECTION 1.2
# Factors of production

**In this section:** 1.2.1 the four factors of production and their rewards · 1.2.2 the quantity and quality of factors of production

## Key words

### Factors of production
The resources used to produce goods and services: land, labour, capital and enterprise.

**Simply:** everything a firm needs to make something.

**các yếu tố sản xuất**

### Land
All natural resources: the land itself, and everything on it, under it and in the sea, such as forests, rivers, fish, oil and minerals.

**Simply:** gifts of nature. Reward: **rent**.

**đất đai (tài nguyên thiên nhiên) · tiền thuê**

### Labour
The human effort, physical and mental, used in production.

**Simply:** workers and their work. Reward: **wages**.

**lao động · tiền lương**

### Capital
Human-made goods used to produce other goods and services, such as machines, tools, factories, vehicles and computers.

**Simply:** things made by people to make other things. Reward: **interest**.

**vốn (tư liệu sản xuất: máy móc, nhà xưởng) · tiền lãi**

### Enterprise
The willingness and ability to organise the other factors of production and to take risks, in order to make a profit.

**Simply:** the entrepreneur has the idea, puts it all together and takes the risk. Reward: **profit**.

**khả năng kinh doanh (doanh nhân) · lợi nhuận**

### Quantity of factors
How much of each factor there is: the number of workers, the amount of land, the number of machines.

**Simply:** how many? Example: more people of working age.

**số lượng các yếu tố sản xuất**

### Quality of factors

How productive each factor is: the skills of workers, the fertility of land, the technology in machines.

**Simply:** how good? Example: training makes workers more productive.

**chất lượng các yếu tố sản xuất · năng suất**

### Memory hook: L-L-C-E → R-W-I-P

| | |
|---|---|
| **L** | **Land** → **Rent**. *A café pays rent for a shop in District 1.* |
| **L** | **Labour** → **Wages**. *A factory worker in Bắc Ninh earns $400 a month.* |
| **C** | **Capital** → **Interest**. *A firm pays interest on a bank loan for new machines.* |
| **E** | **Enterprise** → **Profit**. *The owner of a phở chain keeps the profit, but also takes the risk of a loss.* |

Đất đai → tiền thuê · Lao động → tiền lương · Vốn → tiền lãi · Doanh nhân → lợi nhuận

**TRAP**

### Capital is not money

In everyday Vietnamese, **vốn** often means money to start a business. In economics, **capital** means **machines, tools, buildings and vehicles**. Money is not a factor of production, because money cannot make anything by itself. Money is used to **buy** capital.

**Wrong:** "The firm's capital is its $50 million bank loan." **Right:** "The firm used a $50 million loan to buy capital: new machines."

**So why is interest the reward for capital?** Firms usually borrow money to buy machines. The interest they pay is the price of using that capital, so interest is the reward for capital. In the Sao Mai case, the 6% interest on the loan is the reward for the machines the loan paid for.

**TRAP**

### Land is more than soil

**Land** means **all natural resources**, not only đất. Fish in the sea, the river Mekong, coal in Quảng Ninh, oil under the sea and the forests of Tây Nguyên are all land.

## Extra: mobility of labour

**Extra reading.** Mobility is not in your G9 scheme of work for 1.2, so it will not be tested in this exam. It is here because it helps with the Bắc Ninh question (workers moving from Thanh Hóa and Nghệ An).

**Geographical mobility: can a worker move to a new place?**

It is **lower** when: housing in the new city is expensive; the worker has family ties; schools for children are hard to find; the worker does not know about jobs there.

**Occupational mobility: can a worker move to a new job?**

It is **lower** when: the new job needs skills or qualifications the worker does not have; training is expensive or takes a long time.

Training and education **raise** occupational mobility.

**Watch out:** other factors are mobile too. Land is **geographically immobile** (you cannot move a field), but it can be occupationally mobile (a rice field can become a shrimp pond). A tractor can be moved to another farm, but it cannot be used to sew clothes.

## Quantity and quality

| Factor | Quantity rises when… | Quality rises when… |
|---|---|---|
| **Land** | new resources are found (a new oil field); land is reclaimed from the sea | fertilisers and irrigation make farmland more productive |
| **Labour** | the population grows; more women join the workforce; workers come from other countries; the retirement age rises | education, training and better healthcare |
| **Capital** | firms invest in more machines and buildings | new technology (robots, faster computers) |
| **Enterprise** | more people start businesses (for example, it is easier and cheaper to set one up) | business training and experience |

More factors, or better factors, means the economy **can produce more**. In 1.4 you will see this as an **outward shift of the PPC**.

## Paper 1 practice: multiple choice

**5. A baker in Huế buys a delivery van. Which factor of production is the van?**

**A** land
**B** labour
**C** capital
**D** enterprise

**6. What is the reward for land?**

**A** rent
**B** wages
**C** interest
**D** profit

**7. Which would most likely improve the quality of labour?**

**A** higher rents in big cities
**B** lower wages in factories
**C** fewer jobs in farming
**D** free training courses for workers

**8. Which is an example of land as a factor of production?**

**A** money in a bank
**B** fish in the sea
**C** a fishing boat
**D** a fisherman's skills

## Paper 2 practice: structured question

CONTEXT

### An electronics factory in Bắc Ninh

Sao Mai Electronics makes parts for mobile phones in an industrial park in Bắc Ninh. The owners started the firm 10 years ago and took a big risk.

| Sao Mai Electronics (practice figures) | Last year |
|---|---|
| Workers | 8,000, each paid about $400 a month |
| Rent for land in the industrial park | $2 million a year |
| Bank loan to buy new machines | $50 million at 6% interest a year |
| Profit for the owners | $12 million |

The new machines need skilled technicians. The firm wants 2,000 more, but few local workers have the right skills. Sao Mai has opened a training centre, and many young workers move from Thanh Hóa and Nghệ An to work there.

**Define** **(a) Define the term *enterprise*.** **[2]**

**To get 2 marks** you need two ideas: **organising** the other factors and **taking risks**.

1. **Organise:** *Enterprise is the willingness and ability to _____ the other factors of production…*
2. **Risk:** *…and to take _____ in order to make a profit.*

**Explain** **(b) Explain two factor rewards that Sao Mai Electronics pays. Use the context.** **[4]**

**To get 4 marks:** for each reward, name the reward **and** the factor it pays for (1), then use a figure from the context and say who receives it (1).

1. **Reward 1:** _____ *is the reward for* _____.
2. **Context:** *For example, Sao Mai pays* _____ *to* _____.
3. **Reward 2:** *A second reward is* _____, *which is the reward for* _____.
4. **Context:** *For example,* _____.

**Word bank:** wages · rent · interest · profit · labour · land · capital · enterprise

> **Watch out:** profit goes to the **owners**, not the firm's workers. Interest is paid on the loan used to buy **capital**.

**Analyse** **(c) Analyse how training workers could affect the output of Sao Mai Electronics.** **[6]**

**To get 6 marks:** six links from **training** to **output**. Use the words **quality** and **productivity**.

1. **Quality:** *Training improves the _____ of labour: workers gain new skills.*
2. **Productivity:** *So each worker can produce _____ in an hour. Labour productivity rises.*
3. **Capital:** *Skilled technicians can use the new _____ properly, with fewer mistakes and breakdowns.*
4. **Shortage of skills:** *The firm can fill the _____ technician jobs, so machines are not left unused.*
5. **Output:** *As a result, the factory produces _____ phone parts with the same land and machines, and the cost of each part may fall.*
6. **Sales:** *Because each part costs less to make, Sao Mai can offer phone makers _____ prices, win more orders and produce even more.*

> **Watch out:** training changes the **quality** of labour, not the quantity. The firm does not have more workers. It has **better** workers. Keep every step about **Sao Mai**: the question asks about one firm, not the whole economy.

# 1.3 SECTION 1.3 Opportunity cost

**In this section:** 1.3.1 opportunity cost · 1.3.2 how opportunity cost influences the decisions of consumers, workers, producers and governments

## Key words

### Opportunity cost

The next best alternative that is given up (forgone) when a choice is made.

**Simply:** your second choice, the thing you could have had instead.

**chi phí cơ hội**

### Choice

Because resources are scarce, economic agents must choose between different uses of them.

**Simply:** scarcity forces us to choose.

**sự lựa chọn**

### Next best alternative

The option that would have been chosen if the first choice was not possible.

**Simply:** number 2 on your list.

**phương án tốt nhất tiếp theo (bị bỏ qua)**

### Trade-off

Having more of one thing means having less of another.

**Simply:** more phones means less rice.

**sự đánh đổi**

### How to find the opportunity cost in 3 steps

1. **List** the choices, from best to worst.
2. Find what was **chosen** (number 1).
3. The opportunity cost is **number 2 only**: the next best alternative.

**TRAP**

### Opportunity cost is one thing, and it is not the money

Minh has $20. His list: 1 a book, 2 a cinema ticket, 3 a T-shirt. He buys the book.

**Wrong:** "the $20" (that is the price, not the opportunity cost). **Wrong:** "the cinema ticket and the T-shirt" (do not add up all the options).

**Right:** the opportunity cost is the **cinema ticket**, his next best alternative.

## Opportunity cost and decision-making

| Who | Choice | Opportunity cost | How it affects the decision |
|---|---|---|---|
| **Consumer** | Lan spends $30 on a concert ticket. | the new shoes she wanted | She compares the benefit of the concert with the shoes. If the shoes become more important to her, she may not buy the ticket. |
| **Worker** | Nam works 10 hours overtime a week. | 10 hours of study or rest | If overtime pay rises, the benefit of overtime grows compared with what he gives up, so he may work more hours. |
| **Producer** | A factory uses its line to make phones. | the TVs it could have made | If the price of TVs rises, the opportunity cost of making phones rises, so the firm may switch to TVs. |
| **Government** | It spends $200 million on a metro line. | a new hospital | The government must judge which project gives people more benefit. Voters may not agree. |

Economic agents try to make the choice where the benefit is greater than the opportunity cost. When the opportunity cost of a choice rises, people are **less likely** to make that choice.

## Paper 1 practice: multiple choice

> **9. Minh has $20. His choices, from best to worst, are a book, a cinema ticket and a T-shirt. Each costs $20. He buys the book. What is his opportunity cost?**
>
> **A** the $20 he spent
> **B** the T-shirt
> **C** the cinema ticket and the T-shirt
> **D** the cinema ticket

> **10. The government spends $200 million on a new metro line. Its next best use of the money was a new hospital. What is the opportunity cost?**
>
> **A** the $200 million
> **B** the new hospital
> **C** the new metro line
> **D** the taxes people pay

**11. Lan works 3 hours of overtime instead of studying English for 3 hours. What is the opportunity cost of the overtime?**

**A** the 3 hours of English study

**B** the extra wages she earns

**C** the cost of travel to work

**D** the money her employer pays

**12. A farmer uses a field to grow rice. Why does this choice have an opportunity cost?**

**A** Rice sells at a high price.

**B** The farmer must pay wages.

**C** The field cannot also grow vegetables.

**D** The government buys some of the rice.

## Paper 2 practice: structured question

CONTEXT

### Two choices: Hà in Đà Nẵng and Mr Tâm in Cà Mau

**Hà** is 18 and lives in Đà Nẵng. She can go to university for four years. The fees are **$1,500 a year**. Or she can take a full-time job at a beach hotel that pays **$300 a month**. Hà decides to go to university.

**Mr Tâm** has a 2-hectare rice farm in Cà Mau. Salt water now comes onto his fields in the dry season, so his rice harvest is getting smaller. He is thinking of digging a **shrimp pond** on the same land. The pond costs **$3,000** to build. Shrimp sell for a much higher price than rice. He cannot grow rice and shrimp on the same land at the same time.

**Define** **(a) Define the term *opportunity cost*.** **[2]**

**To get 2 marks** you need two ideas: the **next best alternative**, and that it is **given up** when a choice is made.

1. **What:** *Opportunity cost is the* _____...
2. **When:** *...that is* _____ *when a choice is made.*

> **Watch out:** "what you give up" is not enough for 2 marks. Say **next best** alternative.

**Explain** **(b) Explain the opportunity cost of Hà's choice and of Mr Tâm's possible choice. Use the context.** **[4]**

**To get 4 marks:** for each person, name the opportunity cost (1) and explain why it is the next best alternative, using the context (1).

1. **Hà:** *The opportunity cost of going to university is* _____.
2. **Why:** *This is her next best alternative. She gives up* _____ *a year in wages.*
3. **Mr Tâm:** *If he builds the shrimp pond, the opportunity cost is* _____.
4. **Why:** *This is because land is scarce:* _____.

> **Watch out:** the $1,500 fees are a **money cost**. The question asks for the **opportunity cost**: her next best alternative, which is the hotel job and its wages (12 × $300 = $3,600 a year).

**Analyse** **(c) Analyse how opportunity cost could influence Mr Tâm's decision to switch from rice to shrimp.** **[6]**

**To get 6 marks:** six links. Explain what he gives up, how the salt water changes that, and how he compares benefit and opportunity cost.

1. **Scarce land:** *Mr Tâm's land is scarce. He cannot use it for* _____ *at the same time.*
2. **Opportunity cost:** *So if he switches to shrimp, his opportunity cost is* _____.
3. **Salt water:** *Salt water makes his rice harvest smaller, so the rice he gives up is* _____. *The opportunity cost of shrimp* _____.
4. **Benefit:** *Shrimp sell for a higher price, so the benefit of shrimp is* _____ *than the rice he gives up.*
5. **The $3,000:** *The $3,000 for the pond also has an opportunity cost: he could have used it for* _____.
6. **Decision:** *He compares the benefit with the opportunity cost. Because the opportunity cost has fallen and the benefit is high, he is* _____ *to switch.*

# 1.4 SECTION 1.4
# Production possibility curve (PPC) diagrams

**In this section:** 1.4.1 the PPC · 1.4.2 points under, on and beyond a PPC · 1.4.3 movements along a PPC · 1.4.4 shifts of a PPC

## Key words

### Production possibility curve (PPC)
A curve that shows the maximum combinations of two goods (or services) that an economy can produce when all its resources are fully and efficiently used.

**Simply:** the most you can make of two things with what you have.

**đường giới hạn khả năng sản xuất**

### Efficient · inefficient
A point **on** the PPC is efficient: all resources are used. A point **inside** the PPC is inefficient: some resources are unemployed or wasted.

**Simply:** on the line = nothing wasted. Inside = could make more of both.

**hiệu quả · không hiệu quả**

### Unattainable
A point **outside** (beyond) the PPC. The economy cannot produce it with its current resources and technology.

**Simply:** not possible now.

**không thể đạt được**

### Reallocation of resources
Moving resources from making one good to making another. This is a movement **along** the PPC.

**Simply:** more of one good, less of the other.

**phân bổ lại nguồn lực**

### Economic growth
An increase in an economy's productive capacity, shown by an **outward shift** of the PPC.

**Simply:** the economy can now make more of both goods.

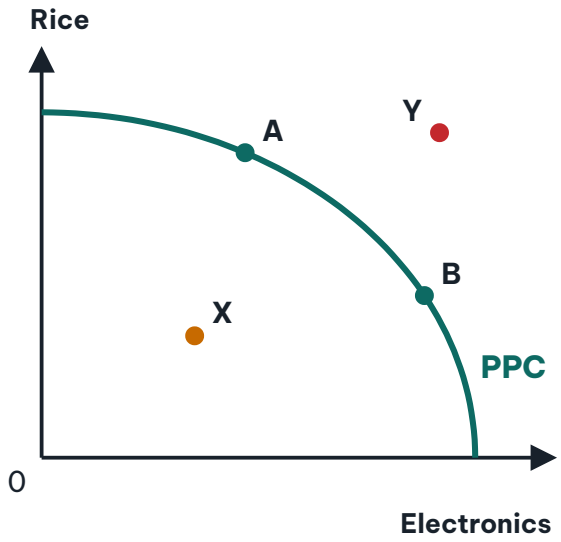
**tăng trưởng kinh tế**

### Inward shift
A fall in productive capacity, because the quantity or quality of resources falls.

**Simply:** the economy can now make less of both goods.

**PPC dịch chuyển vào trong**

## Points on, inside and outside the PPC



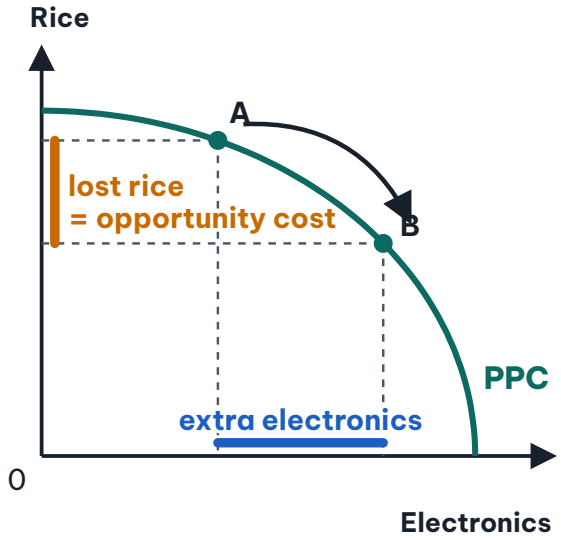

A and B: on the PPC. X: inside. Y: outside.

**A and B (on the curve):** efficient. All resources are used. To make more of one good, the economy must make less of the other.

**X (inside):** inefficient. Some resources are unemployed, for example workers without jobs or closed factories. The economy could make more of **both** goods.

**Y (outside):** unattainable now. The economy needs more resources or better technology to reach it.

## A movement along the PPC shows opportunity cost



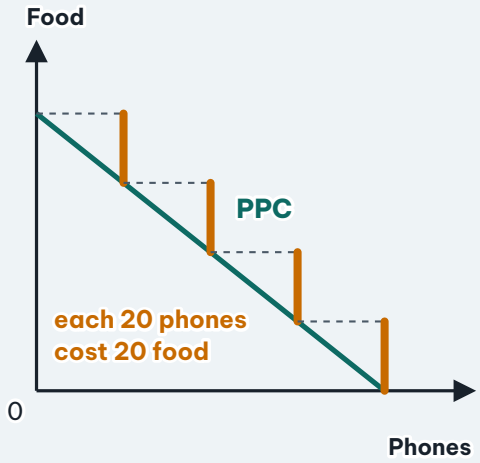

From A to B: more electronics, less rice.

The economy moves resources (workers, land, machines) from rice farming into electronics. It moves **along** the PPC from A to B.

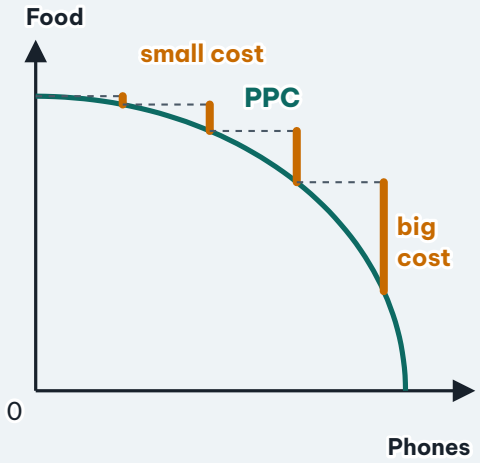
The **gain** is the extra electronics (blue). The **opportunity cost** is the rice given up (orange).

The PPC itself does **not** move. The economy has the same resources. It just uses them differently.

### Why is the PPC curved? Increasing opportunity cost




Straight line: every 20 phones costs the same amount of food. Resources are equally good at making both goods.

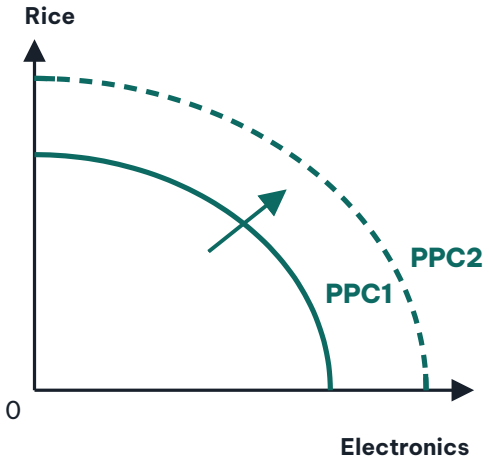



Curved (bowed out): every 20 phones costs more food than the 20 before.

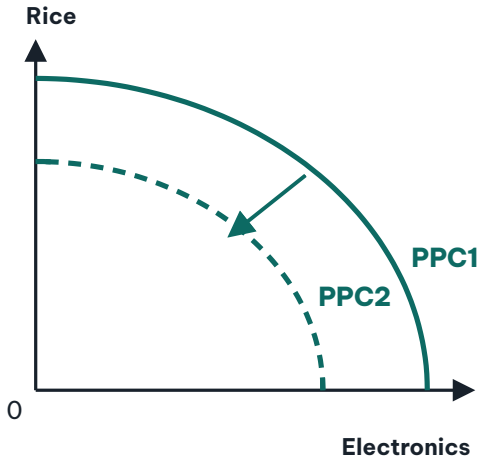
Resources are **not equally good** at making both goods. When a country starts making phones, it first moves the resources that are best at phones and worst at farming, so it loses only a little food. As it makes more and more phones, it must move the best farmland and the best farmers too. Each extra phone now costs **more** food. So the opportunity cost **rises**, and the curve bows out.

PPC có dạng cong lồi ra ngoài vì chi phí cơ hội tăng dần: các nguồn lực không phù hợp như nhau cho việc sản xuất cả hai hàng hóa.

## Shifts of the PPC




Outward shift: the economy can produce more of both goods (economic growth).




Inward shift: the economy can produce less of both goods.

**Outward shift: more or better factors**

- new technology
- more capital (investment in machines, roads, ports)
- better education and training (quality of labour)
- more workers (population growth, immigration)
- new natural resources found

**Inward shift: fewer or worse factors**

- a natural disaster (typhoon, flood, drought) destroys land and capital
- war
- emigration of workers
- a fall in the working population
- natural resources run out

**TRAP**

### Moving from inside the curve to the curve is not a shift

Unemployed workers find jobs. The economy moves from X (inside) **towards the PPC**. The PPC does **not** shift, because the economy has the same resources. It is only using them better.

The PPC shifts only when the **quantity or quality** of resources (or technology) changes.

**TRAP**

### Axis labels

A PPC shows **two goods**. Label the axes with the goods, for example "Rice" and "Electronics" or "Capital goods" and "Consumer goods". **Never** label them Price and Quantity: that is a demand and supply diagram.

## Calculation practice

A country can produce these combinations of food and cars when all resources are used.

| Point | A | B | C | D | E |
|---|---|---|---|---|---|
| **Cars (thousands)** | 0 | 20 | 40 | 60 | 80 |
| **Food (million tonnes)** | 100 | 95 | 85 | 65 | 0 |

1. What is the opportunity cost of moving from B to C? _____
2. What is the opportunity cost of moving from D to E? _____
3. Use your answers to explain why this PPC is bowed out.

   _____
4. The country produces 40 thousand cars and 60 million tonnes of food. Is this point inside, on or outside the PPC? What does it show? _____

## Paper 1 practice: multiple choice

**13. What does a point outside a country's PPC show?**

**A** output with some unemployed resources

**B** output that cannot be reached now

**C** output where all resources are used

**D** output after a fall in productivity

**14. Which would cause an outward shift of a country's PPC?**

**A** more workers become unemployed

**B** a typhoon destroys many factories

**C** workers move from farms to factories

**D** new technology raises productivity

**15. An economy on its PPC moves from 40 cars and 100 tonnes of food to 50 cars and 80 tonnes of food. What is the opportunity cost of the extra 10 cars?**

**A** 10 cars

**B** 80 tonnes of food

**C** 20 tonnes of food

**D** 100 tonnes of food

**16. Why is a PPC usually bowed out (curved)?**

**A** Resources are not equally good at making both goods.

**B** The economy always produces on the curve.

**C** Prices rise when the economy produces more.

**D** The economy has unlimited resources.

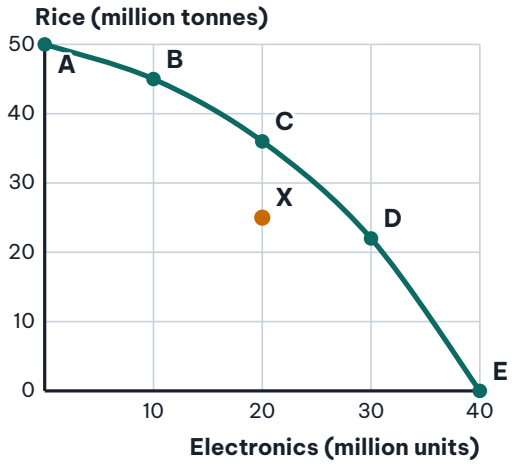
## Paper 2 practice: structured question

CONTEXT

### Việt Nam's PPC: rice and electronics

Việt Nam is one of the world's biggest rice exporters, and it also makes many phones and other electronics. The table and diagram show combinations of rice and electronics that Việt Nam could produce in a year with all its resources (practice figures).

| Point | Electronics (million units) | Rice (million tonnes) |
|---|---|---|
| A | 0 | 50 |
| B | 10 | 45 |
| C | 20 | 36 |
| D | 30 | 22 |
| E | 40 | 0 |




Point X: 20 million units of electronics and 25 million tonnes of rice.

In September 2024, Typhoon Yagi hit northern Việt Nam. Floods covered rice fields, and strong winds damaged factories and machines in Hải Phòng and Quảng Ninh. Some workers were hurt and some had to leave their homes.

**State** **(a) State what point X on the diagram shows.** **[2]**

**To get 2 marks** you need two ideas: what is happening to **resources**, and what this means for **output**.

1. **Resources:** *Point X is inside the PPC, so some resources are* _____.
2. **Output:** *The economy is* _____. *It could produce more of* _____.

**Explain** **(b) Using the table, explain the opportunity cost of producing more electronics.** **[4]**

**To get 4 marks:** say what the opportunity cost is (1); calculate it from the table (1); show that it rises (1); explain why it rises (1).

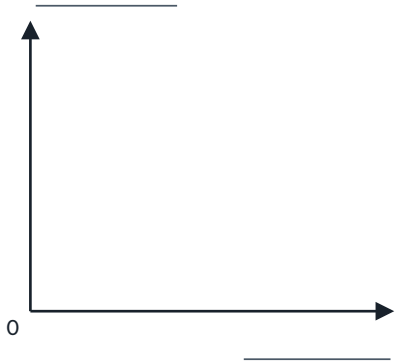
1. **What is given up:** *The opportunity cost of more electronics is the _____ that is given up.*
2. **Calculate:** *For example, moving from B to C, the extra 10 million units cost _____ million tonnes of rice.*
3. **It rises:** *From C to D it costs _____, and from D to E it costs _____. The opportunity cost rises.*
4. **Why:** *This is because resources are not _____. Good farmland and farmers are moved into factories, where they are less productive.*

**Analyse**

## (c) Using a PPC diagram, analyse the effect of Typhoon Yagi on Việt Nam's PPC. [6]

**To get 6 marks:** a labelled PPC diagram (up to **3 marks**) plus a written chain (up to **4 marks**), max 6 in total. Name the factors that are damaged.

1. **Damage:** *The typhoon damaged _____ (land), factories and machines (_____) and hurt some workers (labour).*
2. **Factors:** *So the _____ of factors of production falls.*
3. **Capacity:** *This means Việt Nam's productive capacity falls: it can produce less _____ and less _____.*
4. **PPC:** *The PPC shifts _____, from PPC1 to PPC2.*
5. **Old points:** *Combinations on the old PPC, such as point C, are now _____.*
6. **How long:** *When fields recover and factories are repaired, the PPC may _____. If factories only closed for a short time, the economy moves to a point _____ the PPC instead of a shift.*




Write Rice and Electronics on the lines. Draw PPC1 and PPC2 touching both axes, and add an arrow.

> **Watch out:** Typhoon Yagi damaged farms **and** factories, so the **whole curve** shifts in. (If only farms were damaged, the curve would move in on the rice axis only.)

# Answers and mark schemes

**Model answer colours:** Knowledge: definitions and facts | Analysis: steps in the chain

## Paper 1 answer key

| Q | Answer | Why |
|---|---|---|
| 1 | **B** | The basic economic problem is unlimited wants and finite resources. More money (A) does not create more resources. |
| 2 | **C** | Sunlight is not scarce and uses no resources. Tap water, bus tickets and state-school lessons all use scarce resources, even when they are free of charge (D). |
| 3 | **A** | Robots or workers is a choice of production method. B is "what", C is "for whom". D is about price, which is not one of the three questions. |
| 4 | **D** | Scarcity exists everywhere because wants grow when incomes grow. Rich countries do have free goods (A), such as air. |
| 5 | **C** | A van is human-made and used to produce a service (delivery), so it is capital. |
| 6 | **A** | Land → rent, labour → wages, capital → interest, enterprise → profit. |
| 7 | **D** | Training gives workers new skills, so each worker can produce more. A, B and C do not make workers more skilled. |
| 8 | **B** | Fish are a natural resource, so they are land. Money (A) is not a factor of production; a boat (C) is capital; skills (D) are labour. |
| 9 | **D** | The opportunity cost is the next best alternative only: the cinema ticket. Not the money (A), and not all the options added together (C). |
| 10 | **B** | The hospital is the next best alternative that is given up. The $200 million is the money cost (A). |
| 11 | **A** | She gives up the 3 hours of English study. The wages (B) are the benefit of overtime, not its cost. |
| 12 | **C** | Land is scarce: the field used for rice cannot be used for vegetables at the same time. |
| 13 | **B** | Outside the PPC = unattainable with current resources. A is inside; C is on the curve. |
| 14 | **D** | Better technology raises productive capacity, so the PPC shifts out. A is a point inside; B shifts it in; C is a movement along. |

| | | |
|---|---|---|
| 15 | **C** | Food falls from 100 to 80 tonnes, so 10 extra cars cost 100 − 80 = 20 tonnes of food. |
| 16 | **A** | Resources are not equally suited to both goods, so the opportunity cost rises as more of one good is made. |

## Calculation practice (1.4)

1. B to C: food falls from 95 to 85, so 20 thousand cars cost **10 million tonnes of food**.
2. D to E: food falls from 65 to 0, so 20 thousand cars cost **65 million tonnes of food**.
3. Each extra 20 thousand cars costs more food: 5, then 10, then 20, then 65 million tonnes. The opportunity cost **rises**, because resources that are good at farming are moved into car factories. Rising opportunity cost makes the PPC bowed out.
4. **Inside** the PPC (on the curve, 40 thousand cars goes with 85 million tonnes). It shows that some resources are unemployed: the country could make more food without making fewer cars.

## 1.1 Fresh water in the Mekong Delta

### 1.1(a) Define scarcity. [2]

| Give a mark for | Mark |
|---|---|
| Wants are unlimited / infinite / never end | **1** |
| Resources are finite / limited / not enough to satisfy all wants | **1** |

Do not accept: "when there is not enough of something" on its own (0), or "when a product is expensive" (0).

**MODEL ANSWER**

Scarcity is when people's wants are unlimited **[1]** but the resources to satisfy those wants are finite. **[1]**

## 1.1(b) Economic good and free good. [4]

| Give a mark for | Mark |
|---|---|
| Economic good: scarce / uses resources / has an opportunity cost | **1** |
| Example from the context: water bought from trucks, cleaned or piped water, water from a reservoir | **1** |
| Free good: not scarce / supply is greater than demand at a zero price / no opportunity cost | **1** |
| Example from the context: river water before salt water intrusion | **1** |

**Accept:** sunlight on the fields, sea air, or rain as it falls on the fields as a free-good example, max 1. Do not accept: rain water that families collect and store, because the context says it runs out, so it is scarce. Also do not accept: "a free good is one you do not pay for" (0), because free-of-charge goods can be economic goods.

**MODEL ANSWER**

An economic good is scarce, so resources are used to produce it and it has an opportunity cost. **[1]** For example, the fresh water families buy from trucks for $5 per 1,000 litres needs trucks, drivers and fuel. **[1]** In contrast, a free good is not scarce, because supply is greater than demand at a zero price, so it has no opportunity cost. **[1]** For example, before the salt water came, river water in the Mekong Delta was a free good: families took as much as they needed. **[1]**

### 1.1(c) Why scarcity means the government must decide what, how and for whom. [6]

| Give a mark for each link in the chain (max 6) | Mark |
|---|---|
| Resources (fresh water, the government budget, land, workers) are finite, but wants are unlimited | **1** |
| So not all wants can be satisfied: the government must choose / allocate resources | **1** |
| **What**, with an example: e.g. build a reservoir or pipes; spend on water or on schools | **1** |
| **How**, with an example: e.g. a treatment plant (machines) or water trucks (more workers) | **1** |
| **For whom**, with an example: e.g. homes and hospitals first, then farms | **1** |
| Each choice has an opportunity cost (money for a reservoir cannot be used for something else), so the government must choose carefully to get the most benefit | **1** |

A "what", "how" or "for whom" link with no example from the context gets 0 for that row. Max 6.

**MODEL ANSWER**

Fresh water and the government's budget are finite, but people's wants for water, schools, roads and hospitals are unlimited. **[1]** So the government cannot satisfy every want, and it must decide how to allocate its scarce resources. **[1]** First, it must decide what to produce, for example whether to build a water reservoir or new pipes. **[1]** Second, it must decide how to produce, for example a treatment plant with machines that cleans salty water, or more trucks and drivers to bring water in. **[1]** Third, it must decide for whom, for example whether homes and hospitals get the water before rice farms in the dry season. **[1]** Every choice has an opportunity cost: the money spent on a reservoir cannot be spent on a new school, so the government must choose the use that gives the most benefit. **[1]**

## 1.2 An electronics factory in Bắc Ninh

### 1.2(a) Define enterprise. [2]

| Give a mark for | Mark |
|---|---|
| Organising / combining the other factors of production (land, labour, capital) | 1 |
| Taking risks (to make a profit) | 1 |

Do not accept: "a business" or "a company" (0). Enterprise is a factor of production, not a firm.

**MODEL ANSWER**

Enterprise is the willingness and ability to organise the other factors of production **[1]** and to take risks in order to make a profit. **[1]**

### 1.2(b) Two factor rewards. [4]

| For each of two rewards, give | Mark |
|---|---|
| Names the reward and the factor it pays for: wages → labour; rent → land; interest → capital; profit → enterprise | 1 + 1 |
| Uses the context: $400 a month to the 8,000 workers; $2 million a year to the owner of the land in the industrial park; 6% interest on the $50 million loan for machines; $12 million profit to the owners for their risk | 1 + 1 |

Do not accept: interest as the reward for "money", or profit as a reward paid to workers.

**MODEL ANSWER**

Wages are the reward for labour. **[1]** Sao Mai pays each of its 8,000 workers about $400 a month for their work. **[1]** Profit is the reward for enterprise. **[1]** The owners took a big risk when they started the firm, and last year they received $12 million in profit. **[1]**

## 1.2(c) How training workers could affect output. [6]

| Give a mark for each link (max 6) | Mark |
|---|---|
| Training improves the quality of labour / workers' skills | **1** |
| Labour productivity rises: more output per worker per hour | **1** |
| Skilled workers use the new machines (capital) better: fewer mistakes, faster work, less waste | **1** |
| The firm can fill the 2,000 skilled jobs, so machines are not left unused | **1** |
| Output rises with the same land and capital / the cost per unit falls | **1** |
| Lower costs let Sao Mai offer lower prices / win more orders from phone makers, so its output rises further | **1** |

**Accept also:** output may fall in the short run while workers are away training; trained workers may leave for other firms. Max 6. Do not accept: "the firm has more workers" (training changes quality, not quantity), or points about Việt Nam's PPC or the whole economy (the question is about one firm).

**MODEL ANSWER**

Training improves the quality of labour, because workers learn the skills they need for the new machines. **[1]** This means labour productivity rises: each worker can produce more phone parts in an hour. **[1]** Skilled technicians can also use the expensive machines properly, so there are fewer mistakes and breakdowns. **[1]** Sao Mai can now fill the 2,000 technician jobs, so its new machines are not left unused. **[1]** As a result, the factory produces more output with the same land and capital, and the cost of each part may fall. **[1]** Because each part now costs less to make, Sao Mai can offer phone makers lower prices, win more orders and expand its output even further. **[1]**

## 1.3 Hà and Mr Tâm

### 1.3(a) Define opportunity cost. [2]

| Give a mark for | Mark |
|---|---|
| The next best alternative | 1 |
| Given up / forgone when a choice is made | 1 |

Do not accept: "the cost of something" or "the money you spend" (0). "What you give up" with no "next best" gets 1.

**MODEL ANSWER**

Opportunity cost is the next best alternative **[1]** that is given up when a choice is made. **[1]**

### 1.3(b) The opportunity cost of Hà's and Mr Tâm's choices. [4]

| Give a mark for | Mark |
|---|---|
| Hà: the opportunity cost is the hotel job | 1 |
| Explained: it is her next best alternative; she gives up the wages ($300 a month / $3,600 a year) and work experience | 1 |
| Mr Tâm: the opportunity cost of shrimp is the rice he could have grown | 1 |
| Explained: land is scarce; the same land cannot grow rice and shrimp at the same time | 1 |

Do not accept: "the $1,500 fees" or "the $3,000 pond" as the opportunity cost. They are money costs, not the next best alternative. ("The wages from the hotel job" is fine.)

**MODEL ANSWER**

The opportunity cost of Hà going to university is the job at the beach hotel. **[1]** This is her next best alternative, so she gives up $300 a month, or $3,600 a year, in wages. **[1]** If Mr Tâm builds a shrimp pond, his opportunity cost is the rice he could have grown. **[1]** This is because his land is scarce: the same 2 hectares cannot be used for rice and shrimp at the same time. **[1]**

## 1.3(c) How opportunity cost could influence Mr Tâm's decision. [6]

| Give a mark for each link (max 6) | Mark |
|---|---|
| Land is scarce: it cannot be used for both rice and shrimp | 1 |
| The opportunity cost of switching to shrimp is the rice given up | 1 |
| Salt water makes the rice harvest smaller, so the opportunity cost of shrimp falls | 1 |
| Shrimp sell for a higher price, so the benefit of shrimp is greater than the rice given up | 1 |
| The $3,000 also has an opportunity cost (e.g. new seeds, a machine, savings, his children's school fees) | 1 |
| He compares the benefit with the opportunity cost: when the benefit is greater, he is more likely to switch | 1 |

**Accept also:** risk: if shrimp get a disease or the price falls, the benefit may be lower than the opportunity cost. Max 6.

**MODEL ANSWER**

Mr Tâm's land is scarce, so he cannot use the same fields for rice and shrimp at the same time. **[1]** If he switches to shrimp, his opportunity cost is the rice he gives up. **[1]** Salt water is making his rice harvest smaller, so the rice he gives up is getting smaller. This means the opportunity cost of shrimp is falling. **[1]** At the same time, shrimp sell for a much higher price than rice, so the benefit from shrimp is greater than the rice he gives up. **[1]** However, the $3,000 for the pond also has an opportunity cost, because he could have spent it on new rice seeds or kept it as savings. **[1]** Mr Tâm compares the benefit of shrimp with the opportunity cost. Because the opportunity cost has fallen and the benefit is high, he is more likely to switch to shrimp. **[1]**

# 1.4 Việt Nam's PPC

## 1.4(a) What point X shows. [2]

| Give a mark for | Mark |
|---|---|
| Some resources are unemployed / not fully used / wasted | 1 |
| The economy is inefficient / could produce more of both goods (more rice without less electronics) | 1 |

Do not accept: "X is unattainable" (that is a point outside).

**MODEL ANSWER**

Point X is inside the PPC, so some of Việt Nam's resources, such as workers or factories, are unemployed. [1] The economy is inefficient: it could produce more rice and more electronics. [1]

## 1.4(b) The opportunity cost of producing more electronics. [4]

| Give a mark for | Mark |
|---|---|
| The opportunity cost is the rice given up | 1 |
| A correct calculation from the table: A→B 5, B→C 9, C→D 14 or D→E 22 million tonnes for each extra 10 million units | 1 |
| The opportunity cost rises as more electronics are produced (two figures compared) | 1 |
| Why: resources are not equally suited to both goods (farmland and farmers are less productive in electronics) | 1 |

**MODEL ANSWER**

The opportunity cost of producing more electronics is the rice that Việt Nam gives up. [1] For example, moving from B to C, the extra 10 million units of electronics cost 45 − 36 = 9 million tonnes of rice. [1] The opportunity cost rises: from C to D the next 10 million units cost 14 million tonnes, and from D to E they cost 22 million tonnes. [1] This is because resources are not equally good at making both goods. The best farmland and the most skilled farmers are moved last, so each extra unit of electronics costs more rice. [1]

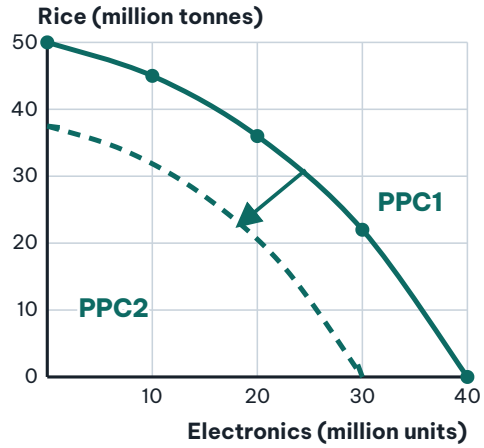
## 1.4(c) The effect of Typhoon Yagi on Việt Nam's PPC. [6]

| Diagram (max 3) | Mark |
|---|---|
| Both axes labelled with the two goods (Rice and Electronics), not Price and Quantity | **1** |
| PPC1 drawn bowed out, touching both axes | **1** |
| PPC2 drawn inside PPC1, touching both axes, with an arrow (or labels) showing the inward shift | **1** |

| Written analysis (max 4) | Mark |
|---|---|
| Factors of production are destroyed or damaged: rice fields (land), factories and machines (capital), workers hurt or moved (labour) | **1** |
| The quantity (and quality) of factors falls, so productive capacity falls: less rice and fewer electronics can be produced | **1** |
| So the PPC shifts inwards / combinations on the old PPC (e.g. point C) are now unattainable | **1** |
| Time: the shift may be temporary, as land recovers and factories are repaired; or, if factories only closed and resources were not destroyed, the economy moves inside the PPC rather than the PPC shifting | **1** |

**Total: max 6.** A student with a full diagram (3) needs three written marks for 6/6. A student with no diagram can earn at most 4.

**Accept:** a shift that is bigger on the rice axis, if explained (more damage to farms). Do not accept: "the economy moves along the PPC" (0).




The typhoon shifts the PPC inwards, from PPC1 to PPC2.

**MODEL ANSWER**

*Diagram as above: axes labelled Rice and Electronics, PPC1, and PPC2 inside it with an arrow.* **[3]**

Typhoon Yagi flooded rice fields (land), damaged factories and machines (capital) and hurt some workers (labour). **[1]** So the quantity of factors of production fell, and Việt Nam can now produce less rice and fewer electronics than before. **[1]** As a result, the PPC shifts inwards, from PPC1 to PPC2, and combinations such as point C are now unattainable. **[1]** The shift may be temporary: when the floods go down and factories are repaired, the PPC can shift back out. **[1]**

*Total: 3 + 4 = 7, capped at the maximum of 6.*

BACK PAGES

# Glossary

Bảng thuật ngữ Anh–Việt

| English | Tiếng Việt | Meaning |
|---|---|---|
| basic economic problem | vấn đề kinh tế cơ bản | using finite resources to satisfy unlimited wants |
| scarcity | sự khan hiếm | unlimited wants, finite resources |
| finite resources | nguồn lực hữu hạn | resources that are limited in amount |
| unlimited wants | nhu cầu vô hạn | wants that never end |
| economic good | hàng hóa kinh tế | scarce, has an opportunity cost |
| free good | hàng hóa tự do | not scarce, no opportunity cost |
| what, how, for whom | sản xuất cái gì, như thế nào, cho ai | the three questions of resource allocation |
| factors of production | các yếu tố sản xuất | land, labour, capital, enterprise |
| land · rent | đất đai (tài nguyên thiên nhiên) · tiền thuê | natural resources and their reward |
| labour · wages | lao động · tiền lương | human effort and its reward |
| capital · interest | vốn (máy móc, nhà xưởng) · tiền lãi | human-made goods used in production and their reward |
| enterprise · profit | khả năng kinh doanh (doanh nhân) · lợi nhuận | organising and risk-taking, and its reward |
| geographical / occupational mobility | tính lưu động theo địa lý / nghề nghiệp | ease of moving to a new place / a new job |
| productivity | năng suất | output per worker (or per unit of a factor) |
| opportunity cost | chi phí cơ hội | the next best alternative given up |
| trade-off | sự đánh đổi | more of one thing means less of another |
| production possibility curve (PPC) | đường giới hạn khả năng sản xuất | maximum output of two goods with all resources used |
| efficient / inefficient | hiệu quả / không hiệu quả | on the PPC / inside the PPC |
| unattainable | không thể đạt được | outside the PPC |
| reallocation of resources | phân bổ lại nguồn lực | a movement along the PPC |
| economic growth | tăng trưởng kinh tế | an outward shift of the PPC |